LECTURE ACTIVITY 1: DON'T GET *RENT* OUT OF SHAPE – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

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Learning Objectives:

By the end of this activity, you should be able to

1) Define and identify observational units.
2) Define and identify variables.
3) Classify variables as categorical or quantitative.
4) Extract basic information from simple data visualizations.
5) Manipulate variables within plots.

Key Terms:

- Data Table
- Data Dictionary
- Observational Unit
- Variable (Categorical vs. Quantitative)
- Exploratory Data Analysis
- Barplot/Bargraph
- Histogram
- Scatterplot

Welcome to our first STAT 1110 Lecture Activity! Lecture Activities are designed to give you a more hands-on approach to learning the material in this course. Feel free to work in groups; there is a lot we can learn from one another! Also, no need to turn this in (though I highly encourage you to complete it!)

Here is a quick re-introduction to some key statistical terms and concepts we learned earlier today. We think of **data** as a collection of values recorded on individuals or objects of interest (called **observational units**). These values are often measurements of various attributes/characteristics, which we call **variables**. At the highest level, variables fall into one of two types: **categorical** or **quantitative**. Categorical variables are those whose values are one of a fixed set of categories (called **levels**), and quantitative variables are those whose values are numbers.

More accurately; quantitative variables are those for which it makes "interpretive sense" to perform arithmetic operations. For example, even if we record "month" using 01 for January, 02 for February, etc., the variable "month" does not become categorical - that's because "01" + "02" (i.e. "January" + "February") has no interpretive meaning!

So, when determining whether a variable is categorical or quantitative, ask yourself: "does it make interpretive sense to perform arithmetic operations on this variable?"

Introduction to the Dataset

Zillow[1] is a popular online database that lists houses and apartments for sale or rent. I (Dr. Ethan) collected the following data from Zillow in late June of this year (2026), by searching for a handful of apartment rentals in San Luis Obispo, CA. Each listing corresponds to a unique apartment, and only apartments were included (i.e. houses and townhomes were excluded from the dataset.) The variables included in this particular dataset are:

[1] https://www.zillow.com/

- `Price`: The rental price (in USD per month)
- `NumBed`: The number of bedrooms (Studio apartments are listed as having "0" bedrooms)
- `NumBath`: The number of baths (including half-baths)
- `Pool`: Whether or not the property has a pool (1 = has a pool, 0 = does not have a pool)
- `Size`: The size (in square feet)

Recall that a specification of all variables along with a short description of each (like what we have above) is what we refer to as a **Data Dictionary**.

Question 1

Based on the data dictionary, classify each of the variables as either quantitative or categorical.

- `Price`: ◯ Categorical √ **Quantitative**
- `NumBed`: ◯ Categorical √ **Quantitative**
- `NumBath`: ◯ Categorical √ **Quantitative**
- `Pool`: √ **Categorical** ◯ Quantitative
- `Size`: ◯ Categorical √ **Quantitative**

Here are the first few rows of the **Data Table**:

Price	NumBed	NumBath	Pool	Size
599	1	1	1	185
599	1	1	1	93
799	1	1	1	122
899	1	1	1	122
899	1	1	1	185
949	1	1	1	185
The table continues, and contains 82 rows in total				

Table 1: First Few Rows of the Data Table

Question 2

What are the observational units in the dataset? How many observational units are there? How can you tell?

Solution: The observational units are **apartment units**, since the data represents values of attributes (like Price, Number of Bedrooms, etc.) associated with individual apartment units. There are **82 observational units in total**, since there are 82 rows in the Data Table and each row of the Data Table corresponds to a unique observational unit.

Exploratory Data Analysis

Many statistics project begins with what is known as **exploratory data analysis (EDA)**. As the name suggests, EDA is dedicated to exploring the dataset before diving into more complex analyses. Indeed, a big part of EDA is generating informative **visualizations** (i.e. **plots**).

Recall that the two types of variables (categorical and quantitative) lend themselves more naturally to different types of plots.

- Categorical variables lend themselves to **barplots/bargraphs**, in which we have one bar for each level and the height of each bar is proportional to the frequency of the corresponding level
- Quantitative variables lend themselves to **histograms**, in which we see how many observations fall into a set of **bins**, each of a fixed (and predetermined) **binwidth**.
  - Quantitative variables with a small number of observations also lend themselves to **dotplots**. We'll talk about the differences between histograms and dotplots a little later.

Well-crafted visualizations can tell us a lot about data! Let's get some practice extracting information from graphs. (We'll talk more about *creating* graphs next lecture.) First, let's get a sense of the **distribution** of properties that have/don't have pools. Based on the classification of the "Pool" variable, we know to make a bargraph:

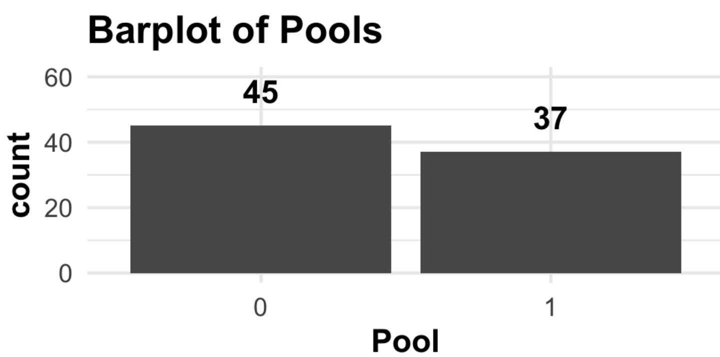

Figure 1: Bargraph of the distribution of which apartments in the dataset have pools and which do not.

Question 3
Which are more common in the dataset: apartment complexes with pools, or apartment complexes without pools? By how much do the counts of these complexes differ - is one type of complex drastically more represented than the other? Explain briefly.

Solution: Apartments without pools are more common than those without pools, as indicated by the fact that there are 45 apartments without pools and 37 with. The difference in magnitude between these two numbers isn't that significant, meaning there isn't a drastic difference between the counts of apartments with and without pools.

Now, let's turn our attention to the prices of apartments. Since "Price" is a quantitative variable, we can visualize its distribution using a histogram:

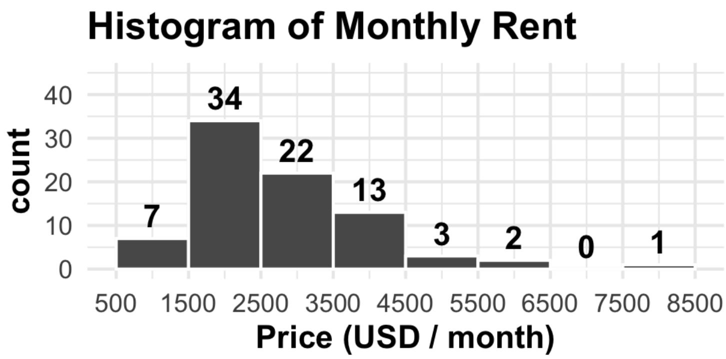

Figure 2: Histogram of the distribution of apartment prices (monthly rents).

Question 4

(a) How many apartments in the dataset have rents between \$500 and \$1500 per month? Or, can this information not be determined from the histogram in Figure (2)?

Solution: There are **7** apartments in the dataset with rents between \$500 and \$1500 per month.

(b) How many apartments in the dataset have rents between \$1500 and \$2500 per month? Or, can this information not be determined from the histogram in Figure (2)?

Solution: There are 34 apartments in the dataset with rents between \$500 and \$1500 per month.

(c) How many apartments in the dataset have rents exactly equal to \$1200 per month? Or, can this information not be determined from the histogram in Figure (2)?

Solution: This information cannot be determined from the histogram; the histogram only gives us the counts of apartments with rents falling in certain ranges, not the exact rents of each apartment.

(d) What is the most common price range, as determined from the histogram in Figure (2)? What about the least common?

Solution: The most common is \$1500-2500 (tallest bar on the histogram), with the least common being \$7500-\$8000 (shortest bar on the histogram). Note that we don't count the range \$6500 - 7500 as being least common, because there were no apartments with rents in that range; customarily, "least common" means "least common that was actually represented in the dataset."

Next, let's explore the distribution of the number of bedrooms. Because **NumBed** is quantitative, we use a histogram:

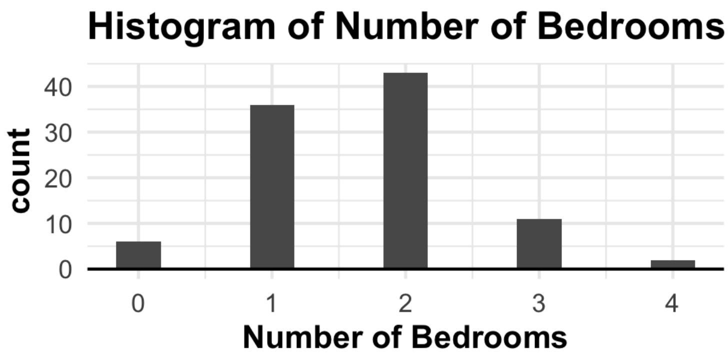

Figure 3: Histogram of the distribution of the number of bedrooms.

Question 5

Why does this histogram have "gaps?" More specifically, what about the **NumBed** variable is qualitatively different from, say, the **Price** variable we explored in the previous question? **Hint:** think about the possible values the **NumBed** variable could take, and compare that with the possible values the **Price** variable could take.

Solution: The key is to note that **NumBed** is constrained to be an integer value - after all, we cannot have a fractional number of bedrooms in an apartment!

We'll expand upon this point a bit more next lecture (and talk about how a different type of plot, called a **dotplot** could have been utilized here). For now, even though the histogram in Figure (3) looks a little different, we can still interpret it like any histogram.

Question 6

(a) What is the most frequently occurring number of bedrooms? Approximately how many apartments in the dataset have this number of bedrooms?

Solution: The most frequently occurring number of bedrooms is **2**, since the bar corresponding to the value of 2 is the tallest out of all bars on the histogram. There are approximately **42 or 43** 2-bedroom apartments included in the dataset, as indicated by the height of the bar.

(b) What is the least frequently occurring number of bedrooms? Approximately how many apartments in the dataset have this number of bedrooms?

Solution: The lease frequently occurring number of bedrooms is **4**, since the bar corresponding to the value of 4 is the shortest out of all bars on the histogram. There are approximately **1 or 2** 2-bedroom apartments included in the dataset, as indicated by the height of the bar.

More Involved Plots

Now that we have a feel for extracting information from histograms and bargraphs, let's take this a bit further! Specifically, we'll work toward answering the following questions through a single plot:

- What, if any, is the relationship between the size of an SLO-area apartment and its rental price?
- Does whether or not the apartment have a pool affect this relationship?

- How, if at all, does the number of bedrooms in the apartment affect this relationship?

First, let's focus on answering the first question: "what, if any, is the relationship between the size of an SLO-area apartment and its rental price?" This is a question about the relationship between the **Size** and **Price** variables, which is a comparison between two quantitative variables. As such, a **scatterplot** is best suited to answer this question.

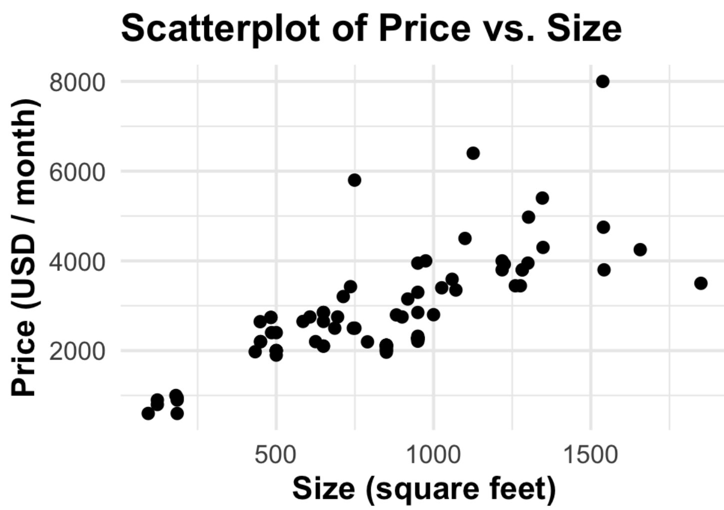

Figure 4: Scatterplot comparing the size and prices of apartments included in the dataset.

Question 7

What (if any) is the relationship between an apartment's size and its rent? Are larger apartments more or less expensive than smaller apartments? Is the relationship (if it exists) linear, or nonlinear - that is, do the points seem to fall along a line, or do they more closely fall along a curve?

Solution: The points seem to fall along a line with positive slope. The "positive slope" means, as size increases so too does price - i.e. larger apartments tend to be more expensive than smaller ones. The fact that the points seem to roughly fall along a line indicates that this relationship is linear.

Note: we typically say things like "tend to" or "on average," to indicate the fact that there are certain exceptions to the rule. For example, there are some individual 1000 ft^2 apartments that are cheaper than 500 ft^2 apartments, but those are the exception rather than the rule. The majority of points display an upward trend, meaning larger apartments tend to be more expensive than smaller ones.

Hm, I wonder if the relationship between an apartment's price and its square footage changes depending on whether or not the apartment has a pool? Note that each point in Figure 4 above represents an individual apartment. As such, we can **change the shape of each point depending on whether or not the associated complex has a pool!**

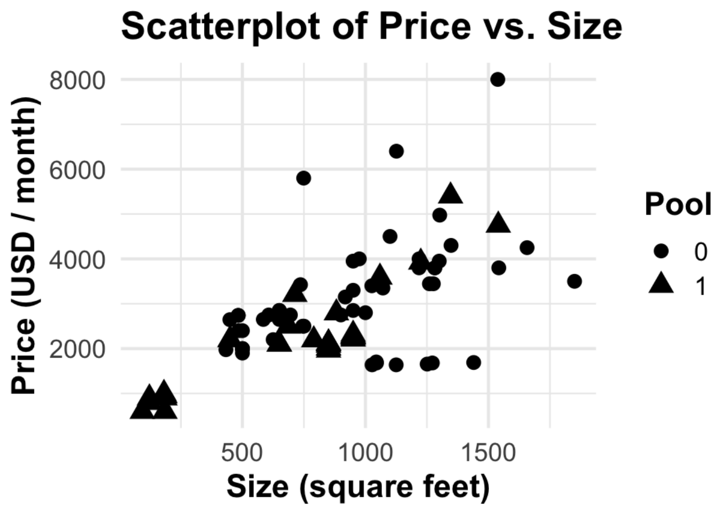

Figure 5: Scatterplot comparing the size and prices of apartments included in the dataset, with each point's shape corresponding to whether or not the apartment has a pool.

Question 8

Does the relationship you described in Question (7) above change depending on whether or not the associated apartment complex has a pool? Furthermore, are apartments with pools significantly more expensive than those without? Are they significantly bigger? How can you tell?

Solution: The key is to try and focus on the circles and triangles separately, as this separates apartments with pools (the circles) from those without pools (the triangles). For both circles and triangles the relationship is still positive; i.e. larger apartments are more expensive than smaller ones.

To answer the second set of questions, note that the circles and triangles overlap significantly; that is to say, there doesn't seem to be a definitive relationship between the prices of apartments with pools and those without pools. If such a relationship existed, we would expect to see a clear separation of circles and triangles in Figure 1 (at least in terms of vertical position).

Finally, we ask the same question about the number of bedrooms. In addition to changing the shape of each point depending on whether or not the associated complex has a pool, we can **change the color of each point to reflect the number of bedrooms.**

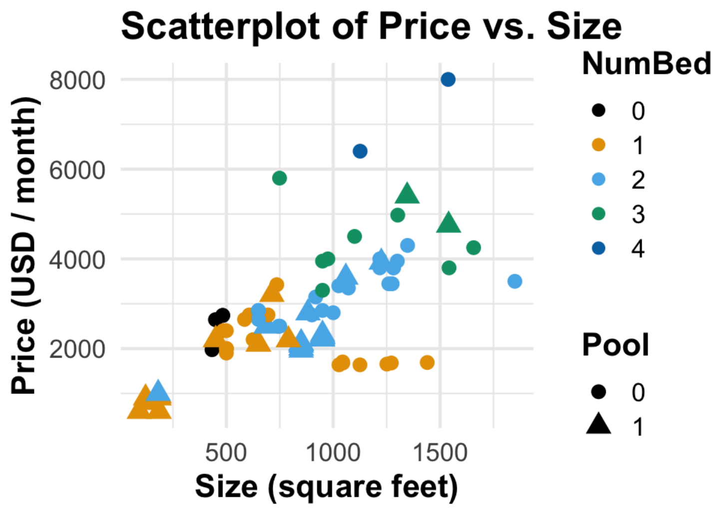

Figure 6: Scatterplot comparing the size and prices of apartments included in the dataset, with each point's shape corresponding to whether or not the apartment has a pool and each point's color corresponding to the number of bedrooms.

Question 9

Describe the relationship between the four variables pictured in Figure (6). Consider questions like: "do bigger apartments tend to have more bedrooms?" and "among apartments with pools, are bigger apartments more expensive than smaller ones?"

Solution: There are only two dark blue points (4 bedroom apartments) so it's a bit difficult to tell how price and size compare among them. (Neither of them have a pool so it's impossible to determine the relationship between having and not having a pool between them.)

Note that, among the green points (3 bedroom apartments) there seems to be a downward slope in the points; i.e. larger 3 bedroom apartments seem to be (on average) cheaper than smaller 3 bedroom apartments. Among all other points, the relationship is still positive.

Once again there isn't a clear separation in terms of having or not having a pool; this lack of separation persists within all color groups (with the possible exception of 1bd apartments, in which apartments with pools seem to be, on average, cheaper than those without).

It should be noted that the number of apartments within each bedroom category is quite small, so it's a little difficult to tell for certain. This question of sample size is one we'll return to in Wednesday's lecture.

On Wednesday, we'll talk more about plots and good practices when it comes to generating them. One thing I'll leave you off with now is: take another look at Figure (6). It contains a lot of information - in fact, I might even say it borders on containing too much information. This is something I'll be stressing a lot throughout this course: oftentimes, simplicity is key. It's important to take care not to overwhelm the reader with any one plot. Again, more about this on Wednesday!

For now, I'd like to close out with one more question.

Question 10

Consider the following histogram of the **NumBath** variable (which, you may recall, tracks the number of bathrooms in each apartment).

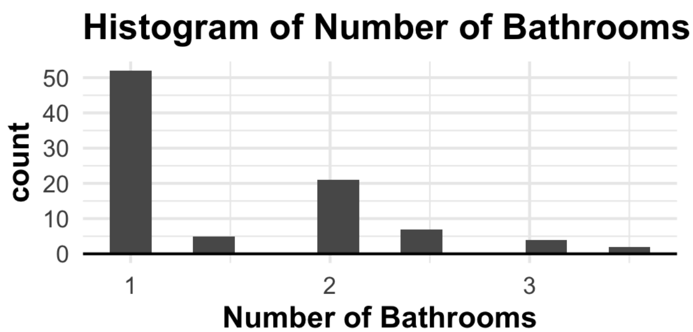

Figure 7: Histogram of the number of bathrooms. Note the presence of half-baths.

A (fictional) student has written the following interpretation.

> The histogram reveals that the most common number of bathrooms is 1, followed by 2, and 3. The presence of half-baths is somewhat rare. This is a phenomenon we can conclude must hold in apartments across the United States.

Do you agree with the student's reasoning? In particular, if you disagree, what part specifically do you disagree with?

Solution: Though the first two lines of the student's work is true (it correctly interprets the plot in Figure (7) above), their final line is not quite correct. In particular, the dataset we've been exploring was collected only on **San Luis Obispo** apartments - we do not know whether the SLO housing market is necessarily reflective of the housing market in the United States as a whole!

This is related to something called **generalizability**, which we will return to frequently throughout this course.